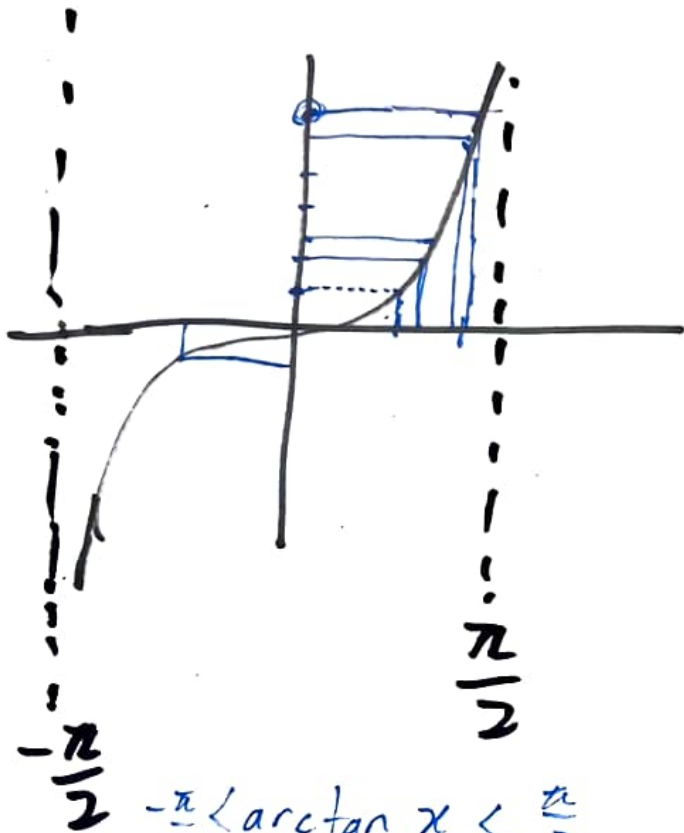


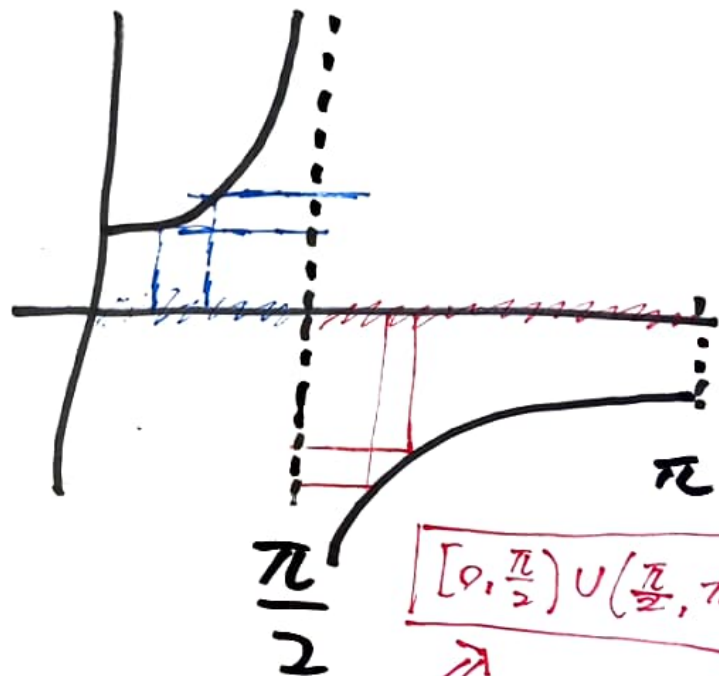
$\tan x$



$$-\frac{\pi}{2} < \arctan x < \frac{\pi}{2}$$

$$n \in \mathbb{N} \Rightarrow 0 < \arctan n < \frac{\pi}{2}$$

$\sec x$



$$[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$$

$$0 \leq \operatorname{arcsec} x < \frac{\pi}{2}$$

or

$$\frac{\pi}{2} < \operatorname{arcsec} x \leq \pi$$

Ex 9.4 Q47

$$-\frac{\pi}{2} < \tan^{-1} n < \frac{\pi}{2}$$

$$\sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n^{1.1}}$$

$$\rightarrow 0 < \frac{\tan^{-1} n}{n^{1.1}} < \frac{\pi/2}{n^{1.1}}$$

→ Can't work

Note that $0 < \tan^{-1} n < \frac{\pi}{2}$ for $n \geq 1$.

$$\Rightarrow 0 < \frac{\tan^{-1} n}{n^{1.1}} < \frac{\pi}{2} \cdot \frac{1}{n^{1.1}}$$

Since $1.1 > 1$, $\sum \frac{1}{n^{1.1}}$ conv.

$$\Rightarrow \sum \frac{\pi}{2} \cdot \frac{1}{n^{1.1}} \text{ conv.}$$

By DCT, $\sum \frac{\tan^{-1} n}{n^{1.1}}$ conv.

Q48 $\sum \frac{\operatorname{arccsc} n}{n^{1.3}} \rightarrow \boxed{0 \leq \operatorname{arccsc} n} < \frac{\pi}{2}$
 $\underbrace{\hspace{10em}}_{n^{1.3}} \quad \underbrace{\hspace{10em}}_{\frac{\pi}{2}}$
 $\underbrace{\hspace{10em}}_{n^{1.3}}$
 conv. by DCT

Q19 $\sum \frac{\sin^2 n}{2^n} \rightarrow \boxed{0 \leq \sin^2 n} \leq \boxed{\frac{1}{2^n}}$
 $\underbrace{\hspace{10em}}_{2^n} \quad \underbrace{\hspace{10em}}_{\frac{1}{2^n}}$
 conv. by DCT $\sum \frac{1}{n^{1.3}}$ conv.
 Conv. Geom. Srs.

Ex 9.5 Q23

$$\sum \frac{2 + (-1)^n}{(1.25)^n}$$

$$\lim(a_n + b_n) = \lim a_n + \lim b_n$$

① Geom. Srs.

$$\begin{aligned}\sum \frac{2 + (-1)^n}{1.25^n} &= \sum \frac{2}{1.25^n} + \sum \left(-\frac{1}{1.25}\right)^n \\ &= 2 \sum \left(\frac{1}{1.25}\right)^n + \sum \left(-\frac{1}{1.25}\right)^n\end{aligned}$$

$\left| \frac{+1}{1.25} \right| < 1$

Df. Since $|\pm \frac{1}{1.25}| < 1$

$\Rightarrow \sum \left(\pm \frac{1}{1.25}\right)^n$ are conv.

$\Rightarrow \sum \frac{2}{1.25^n}$ is conv. $\rightarrow \sum \left(-\frac{1}{1.25}\right)^n$ conv.

$$\sum a_n = \sum \frac{2}{1.25^n} + \sum \left(-\frac{1}{1.25}\right)^n$$

② (Root)

$$\sum \frac{2+(-1)^n}{1.25^n}$$

$$a_n = \frac{2+(-1)^n}{1.25^n}$$

$$\sqrt[n]{|a_n|} = \frac{\sqrt[n]{2+(-1)^n}}{1.25}$$

$$\sqrt[n]{2+(-1)^n}$$

↓

$$1 \leq 2+(-1)^n \leq 3 \Rightarrow \underline{1} \leq \sqrt[n]{2+(-1)^n} \leq \boxed{\sqrt[n]{3}}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \sqrt[n]{2+(-1)^n} = 1 \text{ by Sqz. Thm.}$$

$$\lim \sqrt[n]{|a_n|} = \lim \frac{\sqrt[n]{2+(-1)^n}}{1.25} = \frac{1}{1.25}$$



$$\lim |a_n| \neq 0 \Rightarrow \lim a_n \neq 0$$

$$\equiv \lim a_n = 0 \Rightarrow \lim |a_n| = 0 \quad (\text{Contrapositive})$$

Use Def: (Practice)

Ex 9.6
Q42

$$\sum \underbrace{(-1)^n (\sqrt{n^2+n} - n)}_{a_n} \times \frac{\text{Mult. by Conj.}}{\frac{\sqrt{n^2+n} + n}{\sqrt{n^2+n} + n}}$$

$$|a_n| = \frac{n^2+n - n^2}{\sqrt{n^2+n} + n} = \frac{n}{\sqrt{n^2+n} + n} \times \frac{\frac{1}{n}}{\frac{1}{n}}$$

$$= \frac{1}{\sqrt{1+\frac{1}{n}} + 1} \rightarrow \frac{1}{1+1} = \frac{1}{2} \neq 0$$

$$\lim |a_n| = \dots = \frac{1}{2} \neq 0 \Rightarrow \boxed{\lim a_n \neq 0} \Rightarrow \sum a_n \text{ div. by } n\text{-term test.}$$

~~$\hookrightarrow \sum |a_n| \text{ div.}$~~